CIE International A Level Physics (9702)

Topic 13
Gravitational Fields

Newton's Law · Field Strength · Potential · Orbits · Escape Velocity

Revision Booklet

Syllabus 9702 — A Level (A2) — 2025–2027

This booklet covers:

- Newton's Law of Gravitation
- Gravitational Field Strength
- Gravitational Potential
- Circular Orbits and Satellites
- Escape Velocity
- Formula Summary, Worked Examples, Exam Practice & Checklist

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1 Core Concepts and Definitions

1.1 What is a Gravitational Field?

A gravitational field is a region of space in which any object with mass experiences a force. It is a **vector field** — at every point it has both magnitude and direction (always directed towards the centre of mass of the body creating the field).

1.2 Radial vs Uniform Fields

- **Radial field** (e.g. around a planet): field lines point towards the centre; field strength varies with distance. Represented by lines that converge at the centre.
- **Uniform field** (e.g. near Earth's surface over small distances): field lines are parallel and equally spaced; g is approximately constant at $9.81\,\mathrm{N\,kg^{-1}}$.

2 Newton's Law of Gravitation

Newton's Law of Gravitation

$$F = \frac{Gm_1m_2}{r^2} \tag{1}$$

F	= gravitational force between the two masses (N)
G	= gravitational constant $= 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
m_1, m_2	= the two point masses or uniform spheres (kg)
r	= separation between centres of mass (m)

2.1 Important Features of Newton's Law

- The force is always **attractive** — gravity never repels.
- It is an **inverse square law**: doubling r reduces F by a factor of 4.
- It applies strictly to point masses, but also works for uniform spheres if r is measured from the centre.
- Newton's third law applies: both masses experience equal and opposite forces.

✓ Exam Tip

If asked to show that gravity obeys an inverse-square law, demonstrate that $F \propto 1/r^2$. This might involve calculating Fr^2 for different values and showing it is constant, or comparing the ratio $F_1/F_2 = r_2^2/r_1^2$.

3 Gravitational Field Strength

Definition: Gravitational Field Strength g

The gravitational field strength at a point is the gravitational force per unit mass acting on a small test mass placed at that point.

$$g = \frac{F}{m} \qquad \text{units: } \mathrm{N\,kg^{-1}}$$

Note: g is a vector quantity (directed towards the centre of mass of the source).

Field Strength Equations

$$g = \frac{F}{m} \qquad \text{and} \qquad g = \frac{GM}{r^2} \tag{2}$$

g = gravitational field strength ($\mathrm{N\,kg^{-1}}$ or $\mathrm{m\,s^{-2}}$)
F = gravitational force on a test mass (N)
m = test mass (kg)
M = mass of body creating the field (kg)
r = distance from centre of M (m)

! Common Mistake

Do not confuse g (field strength, $\mathrm{N\,kg^{-1}}$) with G (the universal gravitational constant, $\mathrm{N\,m^2\,kg^{-2}}$). They are completely different quantities!

4 Gravitational Potential

Definition: Gravitational Potential ϕ

The gravitational potential at a point is the work done per unit mass to move a small test mass from infinity to that point.

$$\phi = \frac{W}{m} \qquad \text{units: } \mathrm{J\,kg^{-1}}$$

Gravitational potential is **always negative** (zero at infinity; work is done by gravity as mass moves inwards, so the system loses potential energy).

Gravitational Potential and Potential Energy

$$\phi = -\frac{GM}{r} \qquad E_p = m\phi = -\frac{GMm}{r} \tag{3}$$

ϕ = gravitational potential at distance r ($\mathrm{J\,kg^{-1}}$)
E_p = gravitational potential energy (J)
G = $6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
M = mass of body creating the field (kg)
r = distance from centre of M (m)

4.1 Relationship Between g and ϕ

The gravitational field strength g is related to the potential ϕ by:

Field Strength as Potential Gradient

$$g = -\frac{\Delta\phi}{\Delta r} \tag{4}$$

g is the **negative gradient** of the potential–distance graph. A steeper gradient means a stronger field.

✓ Exam Tip

On a ϕ vs r graph, the gradient at any point gives $-g$ at that point. The graph curves towards zero as $r \to \infty$, and the steepest part is near the surface where the field is strongest.

4.2 Equipotential Surfaces

An **equipotential surface** is a surface on which the gravitational potential is the same everywhere.

- No work is done moving a mass along an equipotential surface.
- Equipotentials are always **perpendicular** to field lines.
- For a point mass (or uniform sphere), equipotentials are concentric spheres.

5 Circular Orbits and Satellites

For an object of mass m in a circular orbit of radius r around a mass M, the gravitational force provides the centripetal force:

Orbital Mechanics

$$\frac{GMm}{r^2} = \frac{mv^2}{r} = mr\omega^2 = mr\left(\frac{2\pi}{T}\right)^2 \tag{5}$$

$$v = \sqrt{\frac{GM}{r}} \qquad \text{(orbital speed)} \tag{6}$$

$$T^2 = \frac{4\pi^2}{GM}r^3 \qquad \text{(Kepler's Third Law)} \tag{7}$$

Kepler's Third Law

$$T^2 \propto r^3 \tag{8}$$

For all objects orbiting the same central mass M, the square of the orbital period is proportional to the cube of the orbital radius.

$$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3}$$

5.1 Energy in Circular Orbits

Energy	Expression	Sign
Kinetic energy	$E_k = \dfrac{GMm}{2r}$	Positive
Gravitational PE	$E_p = -\dfrac{GMm}{r}$	Negative
Total energy	$E_{\text{total}} = -\dfrac{GMm}{2r}$	Negative

Note: $E_{\text{total}} = \frac{1}{2}E_p$. Total energy is always negative for a bound orbit.

! Common Mistake

When a satellite moves to a **lower** orbit (smaller r): its speed *increases*, its kinetic energy *increases*, but its total energy *decreases* (becomes more negative). This is counterintuitive — the satellite speeds up yet loses total energy because the drop in potential energy outweighs the gain in kinetic energy.

5.2 Geostationary Orbits

Definition: Geostationary Satellite

A geostationary satellite has the following properties:

- Orbital period $T = 24$ hours (same as Earth's rotation)
- Orbits directly above the equator
- Orbits in the same direction as Earth's rotation (west to east)
- Appears stationary relative to a point on Earth's surface
- Orbital radius $\approx 42\,000$ km from Earth's centre
- Used for: telecommunications, weather satellites, GPS

6 Escape Velocity

Definition: Escape Velocity

The **escape velocity** is the minimum speed needed for an object to escape from a gravitational field to infinity, without further propulsion.

Escape Velocity

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}} \tag{9}$$

Derived by setting total energy to zero at infinity:

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = 0 \quad \Longrightarrow \quad v_{\text{esc}} = \sqrt{\frac{2GM}{R}}$$

Note: escape velocity is **independent** of the mass of the escaping object.

✓ **Exam Tip**

The derivation of escape velocity is a common exam request. Set $E_k + E_p = 0$ at the point of just escaping (total energy = 0 at infinity), then solve for v.

7 Formula Summary Sheet

Formula Summary Sheet

Formula	Meaning	Section
$F = \dfrac{Gm_1m_2}{r^2}$	Newton's Law of Gravitation	2
$g = \dfrac{F}{m}$	Field strength (general)	3
$g = \dfrac{GM}{r^2}$	Field strength (radial)	3
$\phi = -\dfrac{GM}{r}$	Gravitational potential	4
$E_p = -\dfrac{GMm}{r}$	Gravitational potential energy	4
$g = -\dfrac{\Delta\phi}{\Delta r}$	Field strength as potential gradient	4
$v = \sqrt{GM/r}$	Orbital speed	5
$T^2 = \dfrac{4\pi^2r^3}{GM}$	Kepler's Third Law	5
$v_{\text{esc}} = \sqrt{2GM/R}$	Escape velocity	6

$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$, $M_\oplus = 5.97 \times 10^{24}\,\text{kg}$, $R_\oplus = 6.37 \times 10^6\,\text{m}$

8 Exam Technique and Problem-Solving Strategy

8.1 5-Step Strategy for Gravitational Field Problems

1. **Identify the type of problem**: is it force, field strength, potential, orbit, or energy?
2. **Draw a diagram**: mark all masses, distances from centre (not surface!), and directions.
3. **Select the correct formula**: radial field (GM/r^2) or near-surface constant g?
4. **Check units and powers of 10**: distances in metres, masses in kg, r always from the centre.
5. **Check sign conventions**: potential and potential energy are always negative; force and field strength magnitudes are positive.

8.2 Most Common Mistakes

- Using the radius of a planet as the surface height rather than measuring from the centre.
- Forgetting the negative sign on gravitational potential ($\phi = -GM/r$).
- Confusing gravitational potential ($\mathrm{J\,kg^{-1}}$) with gravitational potential energy (J).
- In Kepler's Third Law, forgetting to convert T to seconds or r to metres.
- Applying centripetal force equations to objects on a surface (only valid for orbiting objects).

9 Worked Examples

Question 1 Calculate the gravitational field strength at a height of 400 km above Earth's surface. ($M_E = 5.97 \times 10^{24}$ kg, $R_E = 6.37 \times 10^6$ m)

Solution

$r = R_E + h = 6.37 \times 10^6 + 4.00 \times 10^5 = 6.77 \times 10^6\,\mathrm{m}$

$$g = \frac{GM}{r^2} = \frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{(6.77 \times 10^6)^2} = \frac{3.98 \times 10^{14}}{4.58 \times 10^{13}} = 8.69\,\mathrm{N\,kg^{-1}}$$

Note: this is the field strength at the ISS orbit — astronauts are not weightless because gravity is zero!

Question 2 A satellite orbits Earth at a radius of 4.2×10^7 m. Calculate its orbital period.

Solution

$$T^2 = \frac{4\pi^2 r^3}{GM} = \frac{4\pi^2 \times (4.2 \times 10^7)^3}{6.67 \times 10^{-11} \times 5.97 \times 10^{24}} = \frac{4\pi^2 \times 7.41 \times 10^{22}}{3.98 \times 10^{14}} = 7.36 \times 10^9\,\mathrm{s^2}$$

$$T = \sqrt{7.36 \times 10^9} = 8.58 \times 10^4\,\mathrm{s} \approx 23.8\,\mathrm{hours}$$

This is a geostationary orbit ($T \approx 24$ h).

Question 3 A spacecraft of mass 2500 kg is moved from Earth's surface to an altitude of 800 km. Calculate the increase in gravitational potential energy.

Solution

$r_1 = 6.37 \times 10^6$ m, $\quad r_2 = 6.37 \times 10^6 + 8.00 \times 10^5 = 7.17 \times 10^6$ m

$$\Delta E_p = -\frac{GMm}{r_2} - \left(-\frac{GMm}{r_1}\right) = GMm\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$$\Delta E_p = 6.67 \times 10^{-11} \times 5.97 \times 10^{24} \times 2500 \times \left(\frac{1}{6.37 \times 10^6} - \frac{1}{7.17 \times 10^6}\right) = 1.74 \times 10^{10}\,\mathrm{J}$$

10 Practice Exam Questions

Section A — Short Answer Questions

Q1. State Newton's Law of Gravitation. *[2 marks]*

Q2. Define gravitational field strength and state its units. *[2 marks]*

Q3. Explain why gravitational potential is always a negative quantity. *[2 marks]*

Q4. The gravitational field strength at Earth's surface is $9.81\,\mathrm{N\,kg^{-1}}$. Calculate the field strength at a distance of $3R_E$ from Earth's centre, where R_E is Earth's radius. *[2 marks]*

Q5. Two planets, X and Y, orbit the same star. Planet X has an orbital radius twice that of planet Y. Determine the ratio T_X/T_Y. *[3 marks]*

Section B — Longer Structured Questions

Q6. A satellite of mass m orbits a planet of mass M in a circular orbit of radius r.

(a) Show that the orbital speed of the satellite is given by $v = \sqrt{GM/r}$. *[2 marks]*

(b) Show that the period of the orbit satisfies $T^2 = \dfrac{4\pi^2 r^3}{GM}$. *[2 marks]*

(c) The satellite is moved to a lower orbit. Explain what happens to its speed, kinetic energy, and total energy. *[3 marks]*

Q7. The Moon orbits the Earth with a period of 27.3 days at a mean orbital radius of 3.84×10^8 m.

(a) Use this data to calculate the mass of the Earth. *[3 marks]*

(b) Calculate the gravitational potential at the Moon's orbital radius. *[2 marks]*

(c) A 75 kg astronaut travels from Earth's surface to the Moon's orbital radius. Calculate the change in gravitational potential energy. *[3 marks]*

11 Mark Scheme and Answers

Q1. Any two masses exert an attractive force on each other [1]; the force is proportional to the product of their masses and inversely proportional to the square of their separation [1].

Q2. Gravitational field strength is the gravitational force per unit mass acting on a (small test) mass placed at that point [1]; units: $\mathrm{N\,kg^{-1}}$ (or $\mathrm{m\,s^{-2}}$) [1].

Q3. Gravitational potential is defined as zero at infinity [1]; work is done by gravity as mass moves from infinity inward, so the potential decreases below zero — it is always negative [1].

Q4. $g \propto 1/r^2$; at $3R_E$: $g = 9.81/3^2 = 9.81/9 = 1.09\,\mathrm{N\,kg^{-1}}$ [2].

Q5. By Kepler's Third Law: $T^2 \propto r^3$, so $\left(\frac{T_X}{T_Y}\right)^2 = \left(\frac{2r_Y}{r_Y}\right)^3 = 8$ [2]; $\frac{T_X}{T_Y} = \sqrt{8} = 2\sqrt{2} \approx 2.83$ [1].

Q6(a). Gravitational force provides centripetal force: $\frac{GMm}{r^2} = \frac{mv^2}{r}$ [1]; simplify to give $v = \sqrt{GM/r}$ [1].

Q6(b). Substitute $v = 2\pi r/T$ into $v^2 = GM/r$: $\frac{4\pi^2 r^2}{T^2} = \frac{GM}{r}$ [1]; rearrange to $T^2 = \frac{4\pi^2 r^3}{GM}$ [1].

Q6(c). Speed: increases (since $v \propto 1/\sqrt{r}$) [1]; kinetic energy: increases [1]; total energy: decreases (becomes more negative, since $E = -GMm/2r$) [1].

Q7(a). $T = 27.3 \times 24 \times 3600 = 2.36 \times 10^6\,\mathrm{s}$; $M = \frac{4\pi^2 r^3}{GT^2} = \frac{4\pi^2(3.84 \times 10^8)^3}{6.67 \times 10^{-11}(2.36 \times 10^6)^2}$ [1] $= 6.02 \times 10^{24}\,\mathrm{kg}$ [2].

Q7(b). $\phi = -\frac{GM}{r} = -\frac{6.67 \times 10^{-11} \times 6.02 \times 10^{24}}{3.84 \times 10^8} = -1.05 \times 10^6\,\mathrm{J\,kg^{-1}}$ [2].

Q7(c). $\Delta E_p = m\Delta\phi = m(\phi_{\text{Moon orbit}} - \phi_{\text{surface}})$ [1]; $\phi_{\text{surface}} = -GM/R_E = -6.25 \times 10^7\,\mathrm{J\,kg^{-1}}$; $\Delta\phi = -1.05 \times 10^6 - (-6.25 \times 10^7) = 6.14 \times 10^7\,\mathrm{J\,kg^{-1}}$; $\Delta E_p = 75 \times 6.14 \times 10^7 = 4.6 \times 10^9\,\mathrm{J}$ [2].

12 Revision Checklist

Revision Checklist

Learning Objective	Confidence (1–3)
☐ State and apply Newton's Law of Gravitation	
☐ Define gravitational field strength; use $g = F/m$ and $g = GM/r^2$	
☐ Sketch field line and equipotential diagrams	
☐ Define gravitational potential and explain why it is negative	
☐ Use $\phi = -GM/r$ and $E_p = -GMm/r$	
☐ Apply the relationship $g = -\Delta\phi/\Delta r$	
☐ Derive the expression for orbital speed	
☐ State and apply Kepler's Third Law ($T^2 \propto r^3$)	
☐ Analyse energy changes in satellite orbits	
☐ Describe the properties of a geostationary satellite	
☐ Derive and apply the formula for escape velocity	
☐ Interpret graphs of g vs r and ϕ vs r	

Key: 1 = Need more work 2 = Getting there 3 = Confident

Good luck with your revision!

Remember: understanding why *formulae work is more powerful than memorising them. Practise deriving orbital speed, Kepler's Third Law, and escape velocity from first principles.*

CIE International A Level Physics (9702)

Topics 14, 15 & 16
Thermal Physics

Temperature · Ideal Gases · Thermodynamics

Revision Booklet

Syllabus 9702 — A Level (A2) — 2025–2027

This booklet covers:

- Thermal Equilibrium and Temperature Scales
- Specific Heat Capacity and Specific Latent Heat
- The Mole and the Ideal Gas Equation
- Kinetic Theory of Gases
- Internal Energy
- The First Law of Thermodynamics
- Formula Summary, Worked Examples, Exam Practice & Checklist

Contents

1 Core Concepts and Definitions

1.1 What is Temperature?

Temperature is a measure of the **average random kinetic energy** of the molecules in a substance. It is *not* the same as thermal (internal) energy: a large cold object may contain more internal energy than a small hot object, yet have a lower temperature.

Definition: Thermal Equilibrium

Two objects are in **thermal equilibrium** when there is no net transfer of energy between them — they are at the same temperature.

This is the basis of the *zeroth law of thermodynamics*: if objects A and B are each in thermal equilibrium with object C, then A and B are also in thermal equilibrium with each other.

2 Topic 14 — Temperature

2.1 Temperature Scales

The Celsius Scale

The Celsius scale is defined by two fixed points:

- 0 °C — the ice point (pure water–ice equilibrium at 1 atm)
- 100 °C — the steam point (pure water–steam equilibrium at 1 atm)

The Thermodynamic (Kelvin) Scale

The thermodynamic scale does **not** depend on the properties of any particular substance. Its zero point, **absolute zero**, is the temperature at which molecules have the minimum possible kinetic energy. No substance can be cooled below this point.

Temperature Conversion

$$T\,/\,\mathrm{K} = \theta\,/\,{}^\circ\mathrm{C} + 273.15 \tag{10}$$

T = thermodynamic temperature (K)
θ = Celsius temperature (°C)

✓ Exam Tip

Always convert temperatures to kelvin before substituting into gas-law equations. A temperature *difference* $\Delta\theta$ in °C equals the same numerical difference ΔT in K, so conversion is not needed for differences — but is always required for absolute values in $pV = nRT$.

2.2 Thermometric Properties

A **thermometric property** is a physical quantity that varies measurably and reproducibly with temperature, and can therefore be used to measure it.

Thermometric property	Type of thermometer
Volume of a gas at constant pressure	Gas thermometer
Resistance of a metal (e.g. platinum)	Resistance thermometer (PRT)
E.m.f. of a thermocouple	Thermocouple thermometer
Density of a liquid	Liquid-in-glass thermometer

Different thermometers may give slightly different readings between the fixed points because their thermometric properties do not all vary linearly with temperature in identical ways. The thermodynamic scale avoids this problem entirely.

2.3 Specific Heat Capacity

Definition: Specific Heat Capacity c

The **specific heat capacity** of a substance is the energy required to raise the temperature of 1 kg of that substance by 1 K (or 1 °C).

Energy and Temperature Change

$$Q = mc\Delta T \tag{11}$$

Q = energy supplied or removed (J)
m = mass (kg)
c = specific heat capacity ($\mathrm{J\,kg^{-1}\,K^{-1}}$)
ΔT = temperature change (K or °C)

✓ Exam Tip

In calorimetry experiments the energy supplied by an electric heater is $Q = IVt$. Setting $IVt = mc\Delta T$ and solving for c is a standard exam calculation.

2.4 Specific Latent Heat

During a **change of state** the temperature remains constant while energy is supplied or removed. This energy changes the *potential* energy of the molecules (breaking or forming bonds) without changing their kinetic energy.

Definition: Specific Latent Heat L

The **specific latent heat** of a substance is the energy required to change the state of 1 kg of the substance at constant temperature.

Latent Heat

$$Q = mL \tag{12}$$

Q = energy supplied or removed (J)
m = mass (kg)
L = specific latent heat ($\mathrm{J\,kg^{-1}}$)

Two types must be distinguished:

- **Specific latent heat of fusion** L_f — solid ↔ liquid
- **Specific latent heat of vaporisation** L_v — liquid ↔ gas

In general $L_v \gg L_f$ because during vaporisation molecules must be completely separated, requiring far more work against intermolecular forces than during melting.

! Common Mistake

Do not apply $Q = mc\Delta T$ when a substance is changing state. During a change of state the temperature does *not* change — use $Q = mL$ instead. A common error is forgetting to split a calculation into separate stages: warming, change of state, then further warming.

3 Topic 15 — Ideal Gases

3.1 The Mole and Avogadro's Constant

Definition: The Mole

One **mole** of a substance contains exactly $N_A = 6.02 \times 10^{23}$ particles, where N_A is the **Avogadro constant**.

Key relationships:

$$N = nN_A \qquad \text{(number of molecules from moles)} \tag{13}$$

$$n = \frac{m_{\text{sample}}}{M_r} \qquad \text{(moles from mass and molar mass)} \tag{14}$$

3.2 The Ideal Gas Equation

The empirical gas laws (Boyle's, Charles's, Pressure Law) combine into a single equation of state for an ideal gas:

Ideal Gas Equation

$$pV = nRT \tag{15}$$

$$pV = NkT \tag{16}$$

p = pressure (Pa)
V = volume (m^3)
n = amount of substance (mol)
R = molar gas constant = $8.31\,\mathrm{J\,mol^{-1}\,K^{-1}}$
T = thermodynamic temperature (K)
N = number of molecules
k = Boltzmann constant = $1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$

The two constants are related by:

$$k = \frac{R}{N_A} \tag{17}$$

A gas **obeys** the ideal gas equation if $pV \propto T$ — equivalently, if pV/T is constant for a fixed mass of gas.

✓ Exam Tip

For a fixed mass of gas changing conditions:

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

Use $pV = nRT$ when moles are given; use $pV = NkT$ when number of molecules is given.

3.3 Kinetic Theory of Gases

Assumptions of the Kinetic Model

- The gas consists of a **large number** of identical molecules.
- Molecules are in **continuous, random motion**.
- The **volume of the molecules** is negligible compared to the volume of the container.
- **Intermolecular forces** are negligible except during collisions.
- All collisions (molecule–molecule and molecule–wall) are **perfectly elastic**.
- The **duration of a collision** is negligible compared to the time between collisions.

Pressure from Kinetic Theory

Kinetic Theory Equation

$$pV = \tfrac{1}{3} N m \langle c^2 \rangle \tag{18}$$

N = total number of molecules
m = mass of one molecule (kg)
$\langle c^2 \rangle$ = mean-square speed ($\mathrm{m^2\,s^{-2}}$)

The **root-mean-square speed** is defined as:

$$c_{\text{r.m.s.}} = \sqrt{\langle c^2 \rangle} \tag{19}$$

Derivation Sketch

1. Consider one molecule of mass m, speed c_x normal to a wall of area A in a box of length L.
2. Momentum change per collision: $\Delta p = 2mc_x$.
3. Time between successive collisions with the same wall: $\Delta t = 2L/c_x$.
4. Force from one molecule: $F = mc_x^2/L$.
5. For N molecules, extend to 3-D using isotropy $\langle c_x^2 \rangle = \frac{1}{3}\langle c^2 \rangle$:

$$p = \frac{Nm\langle c^2 \rangle}{3V} \quad \Longrightarrow \quad pV = \tfrac{1}{3}Nm\langle c^2 \rangle$$

3.4 Average Kinetic Energy of a Molecule

Comparing $pV = NkT$ with $pV = \frac{1}{3}Nm\langle c^2 \rangle$:

Average Translational Kinetic Energy

$$\langle E_k \rangle = \tfrac{1}{2}m\langle c^2 \rangle = \tfrac{3}{2}kT \tag{20}$$

The average kinetic energy of a molecule depends **only on temperature**.

! Common Mistake

$\langle E_k \rangle = \frac{3}{2}kT$ gives the kinetic energy of **one molecule**. For the total kinetic energy of n moles use $E_{\text{total}} = \frac{3}{2}nRT$. Do not confuse k (Boltzmann constant, per molecule) with R (molar gas constant, per mole).

4 Topic 16 — Thermodynamics

4.1 Internal Energy

Definition: Internal Energy U

The **internal energy** of a system is the sum of the random kinetic and potential energies of all its molecules. It is determined solely by the **state** of the system (its temperature, pressure, volume and composition).

For an **ideal gas** there are no intermolecular forces, so internal energy consists entirely of kinetic energy:

$$U = \tfrac{3}{2}NkT = \tfrac{3}{2}nRT \qquad \text{(ideal monatomic gas)} \tag{21}$$

Hence for an ideal gas, U depends *only* on T.

For **real substances** the potential energy between molecules is also significant. A rise in temperature increases kinetic energy; a change of state at constant temperature changes potential energy only.

4.2 Work Done by a Gas

Work Done at Constant Pressure

$$W = p\,\Delta V \tag{22}$$

W = work done **by** the gas (J)
p = pressure (Pa)
ΔV = change in volume (m^3)

Sign convention: expansion ($\Delta V > 0$) means the gas does positive work on the surroundings; compression ($\Delta V < 0$) means the surroundings do positive work on the gas.

4.3 The First Law of Thermodynamics

Definition: First Law of Thermodynamics

The increase in internal energy of a system equals the energy supplied to the system by heating *plus* the work done *on* the system:

$$\Delta U = q + W \tag{23}$$

Sign conventions (CIE syllabus form $\Delta U = q + W$):

Symbol	Positive when...	Negative when...
ΔU	internal energy increases	internal energy decreases
q	heat flows **into** system	heat flows **out** of system
W	work done **on** system	work done **by** system

✓ Exam Tip

CIE uses $\Delta U = q + W$ where W is work done **on** the gas. Some textbooks write $\Delta U = q - W$ where W is work done **by** the gas. Both are equivalent — always state your sign convention before substituting.

Special Processes

Process	Constraint	Consequence
Isothermal	$\Delta T = 0 \Rightarrow \Delta U = 0$	$q = -W$: heat in equals work out
Adiabatic	$q = 0$	$\Delta U = W$: compression raises U
Isochoric	$\Delta V = 0 \Rightarrow W = 0$	$\Delta U = q$: all heating changes U
Isobaric	$p = \text{const}$	$W = p\Delta V$; both q and W contribute

5 Formula Summary Sheet

Formula Summary Sheet

Formula	Meaning	Topic
$T/\mathrm{K} = \theta/°\mathrm{C} + 273.15$	Temperature conversion	14
$Q = mc\Delta T$	Heat and temperature change	14
$Q = mL$	Latent heat	14
$pV = nRT$	Ideal gas (molar form)	15
$pV = NkT$	Ideal gas (molecular form)	15
$k = R/N_A$	Boltzmann constant	15
$pV = \frac{1}{3}Nm\langle c^2\rangle$	Kinetic theory — pressure	15
$\langle E_k\rangle = \frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$	Average KE per molecule	15
$c_{\mathrm{r.m.s.}} = \sqrt{\langle c^2\rangle}$	Root-mean-square speed	15
$U = \frac{3}{2}NkT = \frac{3}{2}nRT$	Internal energy (ideal gas)	16
$W = p\,\Delta V$	Work done by gas (const. p)	16
$\Delta U = q + W$	First law of thermodynamics	16

6 Exam Technique and Problem-Solving Strategy

6.1 5-Step Strategy for Thermal Physics Problems

1. **Identify the topic**: temperature/heat, ideal gas, or thermodynamics?
2. **List what is given** and what is being asked.
3. **Convert units**: temperatures to K; volumes to m^3; pressures to Pa.
4. **Select the correct formula** and rearrange *before* substituting.
5. **Check your answer**: correct units, sensible order of magnitude, correct sign.

6.2 Key Tips by Topic

Topic 14 — Temperature and Thermal Properties

- Split mixed heating problems into separate stages: warming + change of state + further warming.
- Remember $L_v \gg L_f$: for water, $L_v \approx 2.26 \times 10^6\,\mathrm{J\,kg^{-1}}$ vs $L_f \approx 3.34 \times 10^5\,\mathrm{J\,kg^{-1}}$.
- Calorimetry: set $IVt = mc\Delta T$; watch for heat losses.

Topic 15 — Ideal Gases and Kinetic Theory

- Use $pV = nRT$ for moles; use $pV = NkT$ for number of molecules.
- The kinetic theory derivation is examinable — practise the 3-step argument (momentum change → force → pressure, then extend to 3-D).
- R.m.s. speed $\propto \sqrt{T}$: doubling T in kelvin multiplies $c_{\text{r.m.s.}}$ by $\sqrt{2}$, *not* by 2.

Topic 16 — Thermodynamics

- Be rigorous about sign conventions: state them explicitly before substituting.
- For an isothermal process with an ideal gas, $\Delta U = 0$ always.
- Work done **on** the gas is $+W$ in CIE's first law convention.

7 Worked Examples

Question 1 An electric heater of power 60 W heats 0.50 kg of water from 20 °C to 70 °C in 175 s. Calculate the specific heat capacity of water.

Solution

Energy supplied: $Q = Pt = 60 \times 175 = 1.05 \times 10^4$ J
Using $Q = mc\Delta T$:

$$c = \frac{Q}{m\,\Delta T} = \frac{1.05 \times 10^4}{0.50 \times 50} = 4200\,\text{J kg}^{-1}\,\text{K}^{-1}$$

Question 2 A gas occupies $2.0 \times 10^{-3}\,\text{m}^3$ at a pressure of 1.5×10^5 Pa and temperature 27 °C. Calculate the number of moles of gas.

Solution

Convert: $T = 27 + 273.15 = 300.15 \approx 300\,\text{K}$

$$n = \frac{pV}{RT} = \frac{1.5 \times 10^5 \times 2.0 \times 10^{-3}}{8.31 \times 300} = \frac{300}{2493} = 0.120\,\text{mol}$$

Question 3 Calculate the r.m.s. speed of nitrogen molecules ($M_r = 28\,\text{g mol}^{-1}$) at 20 °C.

Solution

$$T = 293.15\,\text{K}; \quad m = \frac{28 \times 10^{-3}}{6.02 \times 10^{23}} = 4.65 \times 10^{-26}\,\text{kg}$$

$$\langle c^2 \rangle = \frac{3kT}{m} = \frac{3 \times 1.38 \times 10^{-23} \times 293.15}{4.65 \times 10^{-26}} = 2.61 \times 10^5\,\text{m}^2\,\text{s}^{-2}$$

$$c_{\text{r.m.s.}} = \sqrt{2.61 \times 10^5} = 511\,\text{m s}^{-1}$$

Question 4 A gas is compressed. The work done *on* the gas is 450 J and at the same time 200 J of heat is removed from the gas. Find the change in internal energy.

Solution

Work done on gas: $W = +450\,\text{J}$ (positive: on the gas)
Heat removed from gas: $q = -200\,\text{J}$ (negative: leaves system)

$$\Delta U = q + W = -200 + 450 = +250\,\text{J}$$

Internal energy *increases* by 250 J.

8 Practice Exam Questions

Section A — Short Answer Questions

Q1. Define specific heat capacity and state its SI unit. *[2 marks]*

Q2. Distinguish between the specific latent heat of fusion and the specific latent heat of vaporisation. *[2 marks]*

Q3. State *three* assumptions of the kinetic theory of gases. *[3 marks]*

Q4. A 2.0 kg block of ice at 0 °C is melted and the resulting water is heated to 100 °C. Given $L_f = 3.34 \times 10^5\,\mathrm{J\,kg^{-1}}$ and $c_{\text{water}} = 4200\,\mathrm{J\,kg^{-1}\,K^{-1}}$, calculate the total energy required. *[3 marks]*

Q5. The temperature of an ideal gas is doubled on the kelvin scale (at constant volume). State and explain what happens to:

- the r.m.s. speed of the molecules,
- the pressure of the gas.

[4 marks]

Section B — Longer Structured Questions

Q6. An ideal gas is enclosed in a container of fixed volume $5.0 \times 10^{-3}\,\mathrm{m}^3$. The gas contains 3.0×10^{23} molecules at a temperature of 300 K.

(a) Calculate the pressure of the gas. *[2 marks]*

(b) The temperature is raised to 450 K. Calculate the new pressure, assuming the gas remains ideal. *[2 marks]*

(c) Explain, using kinetic theory, why the pressure increases when the temperature is raised. *[3 marks]*

Q7. A sample of gas absorbs 600 J of heat from its surroundings and expands at a constant pressure of 1.2×10^5 Pa, increasing its volume by $2.0 \times 10^{-3}\,\mathrm{m}^3$.

(a) Calculate the work done *by* the gas. *[2 marks]*

(b) Using the first law of thermodynamics, calculate the change in internal energy of the gas. *[3 marks]*

(c) The gas is then compressed adiabatically back to its original volume. State what happens to the internal energy and explain why. *[2 marks]*

Q8. Show that the average translational kinetic energy of a molecule of an ideal gas is $\frac{3}{2}kT$. *[3 marks]*

9 Mark Scheme and Answers

Q1. Specific heat capacity is the energy required to raise the temperature of 1 kg of a substance by 1 K [1]; unit: $\mathrm{J\,kg^{-1}\,K^{-1}}$ [1].

Q2. Latent heat of *fusion*: energy per unit mass to change solid to liquid at constant temperature [1]. Latent heat of *vaporisation*: energy per unit mass to change liquid to gas at constant temperature [1].

Q3. Any three from: large number of molecules [1]; random motion [1]; volume of molecules negligible compared to container [1]; intermolecular forces negligible between collisions [1]; perfectly elastic collisions [1]; collision duration negligible [1]. (3 marks total)

Q4. Melting: $Q_1 = 2.0 \times 3.34 \times 10^5 = 6.68 \times 10^5\,\mathrm{J}$ [1]; heating water: $Q_2 = 2.0 \times 4200 \times 100 = 8.40 \times 10^5\,\mathrm{J}$ [1]; total: $Q = 1.508 \times 10^6\,\mathrm{J} \approx 1.51 \times 10^6\,\mathrm{J}$ [1].

Q5. R.m.s. speed: $c_{\text{r.m.s.}} \propto \sqrt{T}$, so doubling T increases $c_{\text{r.m.s.}}$ by factor $\sqrt{2} \approx 1.41$ [1]; because average KE increases [1]. Pressure: $p \propto T$ at constant V, so pressure doubles [1]; molecules collide with walls more frequently and with greater momentum [1].

Q6(a).

$$p = \frac{NkT}{V} = \frac{3.0 \times 10^{23} \times 1.38 \times 10^{-23} \times 300}{5.0 \times 10^{-3}} = 248\,\mathrm{Pa} \quad [2]$$

Q6(b). $p \propto T$ at constant V: $p_2 = 248 \times \dfrac{450}{300} = 372\,\mathrm{Pa}$ [2].

Q6(c). At higher temperature molecules move faster [1]; they collide with the walls more frequently [1]; each collision transfers greater momentum to the wall, so force and pressure increase [1].

Q7(a). $W_{\text{by gas}} = p\Delta V = 1.2 \times 10^5 \times 2.0 \times 10^{-3} = 240\,\mathrm{J}$ [2].

Q7(b). Work done *on* gas: $W = -240\,\mathrm{J}$ [1]; $\Delta U = q + W = 600 + (-240) = +360\,\mathrm{J}$ [1]; internal energy increases by 360 J [1].

Q7(c). Internal energy increases [1]; no heat enters or leaves (adiabatic), so all work done on the gas during compression increases internal energy [1].

Q8. From kinetic theory: $pV = \frac{1}{3}Nm\langle c^2\rangle$ [1]; from ideal gas: $pV = NkT$; equating and simplifying: $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$ [1]; this equals the average translational kinetic energy of one molecule [1].

10 Revision Checklist

Revision Checklist

Learning Objective	Confidence (1–3)
Topic 14 — Temperature	
☐ Explain thermal equilibrium and state the zeroth law	
☐ Convert temperatures between °C and K using $T = \theta + 273.15$	
☐ State and explain examples of thermometric properties	
☐ Define and use specific heat capacity: $Q = mc\Delta T$	
☐ Define specific latent heat; distinguish L_f and L_v	
☐ Apply $Q = mL$; split calculations across stages correctly	
Topic 15 — Ideal Gases	
☐ Use molar quantities and the Avogadro constant N_A	
☐ Recall and apply $pV = nRT$ and $pV = NkT$	
☐ State the relationship $k = R/N_A$	
☐ State all six assumptions of the kinetic theory	
☐ Reproduce the derivation of $pV = \frac{1}{3}Nm\langle c^2\rangle$	
☐ Derive the result $\langle E_k\rangle = \frac{3}{2}kT$	
☐ Calculate r.m.s. speed from temperature and molar mass	
Topic 16 — Thermodynamics	
☐ Explain internal energy in terms of molecular KE and PE	
☐ State that internal energy of an ideal gas depends only on T	
☐ Calculate work done by/on a gas: $W = p\Delta V$	
☐ Apply the first law $\Delta U = q + W$ with correct sign conventions	
☐ Analyse isothermal, adiabatic, isochoric and isobaric processes	

Key: 1 = Need more work 2 = Getting there 3 = Confident

Good luck with your revision!

Remember: understanding why *formulae work is more powerful than memorising them.*

Practise deriving $pV = \frac{1}{3}Nm\langle c^2\rangle$ *and* $\langle E_k\rangle = \frac{3}{2}kT$ *from first principles.*